Design the Uber API

SystemsExpert design walkthrough

Clarifying questions

- Rider and driver APIs both?
- Payments in scope?
- Real-time location updates required?

Functional requirements

- Rider: request ride, see ETA/driver location, cancel, rate
- Driver: go online/offline, accept ride, update location, complete trip

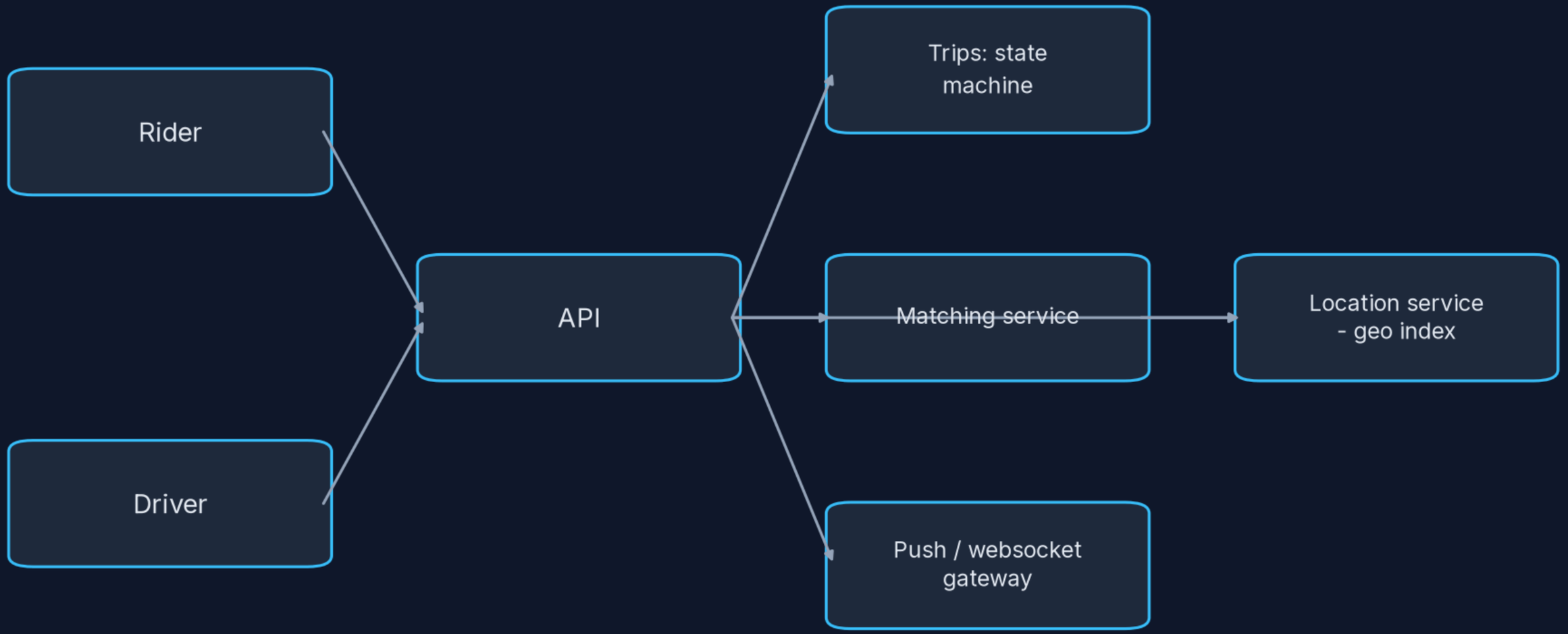
Non-functional requirements

- Real-time updates (~seconds)
- Idempotent state transitions

Capacity estimates

- Drivers send location every ~4 s; ~1M active drivers → ~250k writes/s to the location service

High-level architecture



Data model

- Trip{tripId, riderId, driverId?, status: REQUESTED|ACCEPTED|ARRIVED|IN_PROGRESS|COMPLETED|CANCELLE pickup, dropoff, fare}
- DriverStatus{driverId, online, location, tripId?}

API

```
POST   /trips                          {pickup, dropoff, idempotencyKey} -> Trip
GET    /trips/{tripId}                 -> Trip incl. driver location, ETA
POST   /trips/{tripId}/cancel
POST   /trips/{tripId}/rating          {stars}
PUT    /drivers/me/status              {online: true|false}
PUT    /drivers/me/location            {lat, lng}
POST   /trips/{tripId}/accept
POST   /trips/{tripId}/arrive | /start | /complete
Stream: GET /stream (websocket) -> trip updates, offers to drivers
```

State machine

- Every transition is validated server-side (e.g. accept only from REQUESTED) with optimistic concurrency, so double taps or two drivers accepting resolve deterministically.

Location

- Updates go to an in-memory geo index (geohash/S2 cells) sharded by region; matching queries nearby cells.

Push vs poll

- Websocket for live location; polling fallback.

Trade-offs & alternatives

- Separate endpoints per transition are explicit and auditable vs a generic PATCH status
- In-memory geo index trades durability for speed (locations are ephemeral anyway)

Recap

- Clarify scope, then functional + non-functional needs
- Estimate load to size storage and fan-out
- Core pieces: Rider, API, Driver, Trips, Location service - geo index
- Know your trade-offs and failure modes